

Penalized Regression

David Puelz

The University of Austin

Motivation

Shrinkage: the core idea

Ridge regression

LASSO

Ridge vs LASSO: when to use which?

Choosing λ

Beyond ridge and LASSO

Practice in R



Running example: Texas physicians per capita

Goal: Predict physicians per 10,000 people in a county.

Predictors ($p = 10$): LandArea, PctRural, MedianIncome, Population, PctUnder18, PctOver65, PctPoverty, PctUninsured, PctSomeCollege, PctUnemployed

Questions:

- Do we need all 10? Some may be noise.
- OLS uses all predictors — can we do better by shrinking or dropping some?



The problem with many predictors

When p is large (or p close to n):

- OLS can have **high variance** — coefficients jump around with small data changes
- Risk of **overfitting** — model fits noise, poor out-of-sample performance
- Some predictors may be **irrelevant** — we'd like to ignore them

Two strategies:

1. **Subset selection** — pick which variables to include (discrete)
2. **Shrinkage** — keep all variables but penalize large coefficients (continuous)



Limitations of subset selection

- **Best subset:** 2^p models — with $p = 10$ that's 1,024; with $p = 20$ it's over 1 million
- **Stepwise:** Greedy — adds/drops one at a time, can miss better combinations
- **Instability:** Small change in data $\Rightarrow$ very different selected model

Shrinkage is continuous: we smoothly tune how much to penalize, and the solution changes smoothly with the data.

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Shrinkage in one sentence

Instead of dropping variables, **shrink** their coefficients toward zero.

Why this helps:

- Reduces **variance** — estimates are more stable
- Introduces a little **bias** — we accept being slightly wrong to gain stability
- Often improves **out-of-sample prediction** — the bias-variance tradeoff

The penalty makes it “costly” to have large β_j ; the optimizer balances fit (RSS) vs. penalty.



We are interested in solving a constrained optimization problem!

$$\min_{\beta} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s$$

for some “budget” s . We’re not allowed to spend too much on large coefficients.

LASSO uses $\sum_{j=1}^p |\beta_j| \leq s$ instead — a different “budget” shape that allows some $\beta_j = 0$.

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Ridge regression: the objective

Minimize (over β):

$$\underbrace{\sum_{i=1}^n \left(Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ij} \right)^2}_{\text{RSS}} + \underbrace{\lambda \sum_{j=1}^p \beta_j^2}_{\text{L2 penalty}}$$

for $\lambda \geq 0$. (Typically we do **not** penalize β_0 .)

Intuition: Every unit of β_j^2 costs λ . The optimizer trades off fit vs. penalty.



Ridge: what happens as λ changes?

- $\lambda = 0$: No penalty $\Rightarrow$ ordinary least squares. Full flexibility, possibly high variance.
- λ small: Light penalty. Coefficients shrink a bit; variance drops, bias rises slightly.
- λ large: Strong penalty. Coefficients shrink a lot toward zero; high bias, low variance.
- $\lambda \rightarrow \infty$: All $\hat{\beta}_j \rightarrow 0$ (except intercept). We're saying "ignore all predictors."

The art: Pick λ that minimizes out-of-sample error (e.g., via cross-validation).

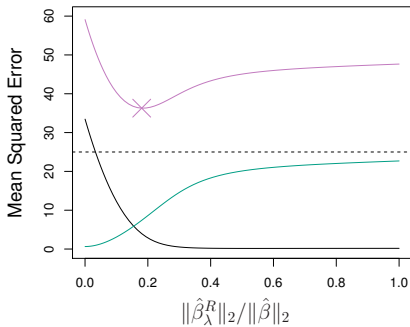
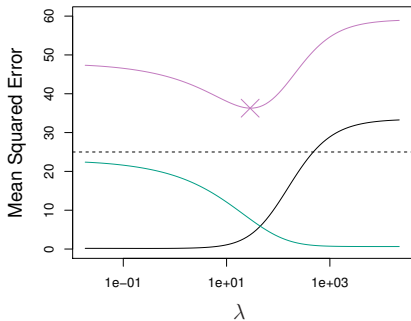


Ridge and the bias-variance tradeoff

- Larger $\lambda \Rightarrow$ more shrinkage $\Rightarrow$ more bias (we're pulling estimates toward zero)
- Larger $\lambda \Rightarrow$ less variance (estimates are more stable)
- Test $MSE = \text{bias}^2 + \text{variance} + \text{irreducible error}$
- Sweet spot: λ where the drop in variance outweighs the rise in bias



Ridge: bias, variance, and test MSE

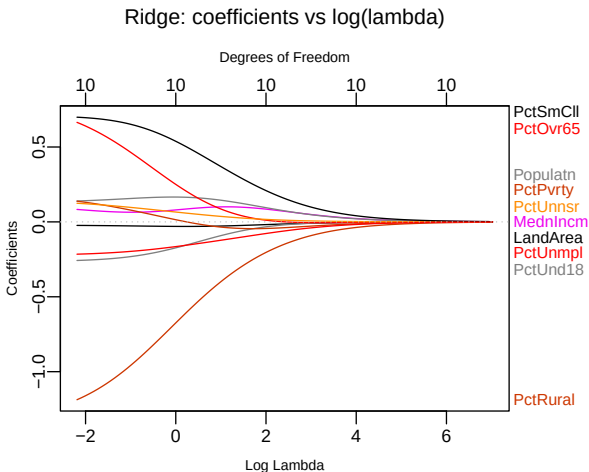


$bias^2$ (black), variance (green), test MSE (purple). Optimal λ minimizes test MSE.



Ridge: coefficient paths (physicians data)

As λ increases, all coefficients shrink toward zero. None become exactly zero.





Why center and scale?

The penalty $\lambda \sum_j \beta_j^2$ treats all coefficients equally. But:

- If X_1 is in dollars and X_2 is a percentage, their scales differ
- A “large” β_1 might be 0.001; a “large” β_2 might be 50
- Without scaling, the penalty would unfairly favor shrinking variables on larger scales

Solution: Center and scale each X_j so they're on comparable footing. `glmnet` does this by default when `standardize=TRUE`.

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LASSO: the key difference

LASSO uses the **L1** penalty instead of L2:

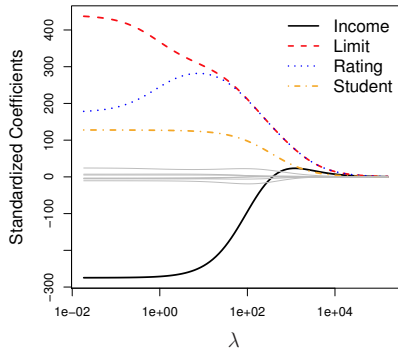
$$\text{RSS} + \lambda \sum_{j=1}^p |\beta_j|$$

Key property: Some $\hat{\beta}_j$ are **exactly zero**. LASSO does automatic variable selection.

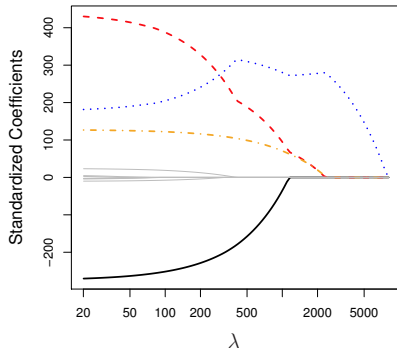
Ridge shrinks but rarely gives exact zeros. LASSO can drop variables entirely.



Ridge vs LASSO: coefficient paths



Ridge: smooth shrinkage



LASSO: some hit zero



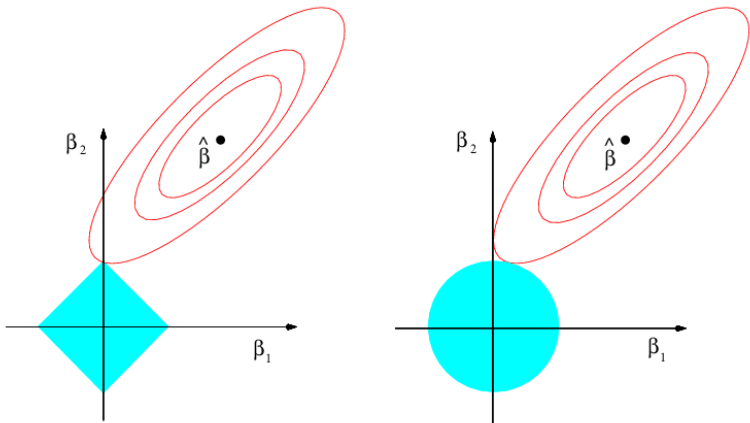
LASSO: the objective

$$\min_{\beta} \left\{ \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

- $|\beta_j|$ is the L1 norm; β_j^2 is the L2 norm (ridge)
- The “kink” at zero in $|\beta_j|$ is what allows coefficients to hit exactly zero
- Again: choose λ by cross-validation



Why does LASSO produce zeros? Geometry



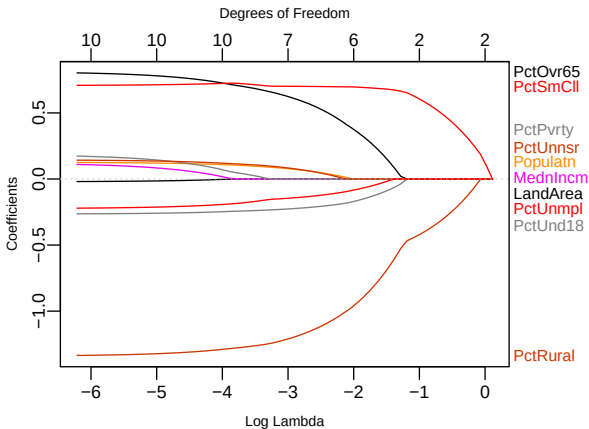
L1 constraint (diamond) has corners on the axes $\Rightarrow$ solution can land at $\beta_j = 0$. L2 (circle) has no corners $\Rightarrow$ solution rarely on axis.



LASSO: coefficient paths (physicians data)

As λ increases, coefficients shrink. Some hit zero and stay there.

LASSO: coefficients vs log(lambda)





How many variables does LASSO keep?

At $\lambda = 0$: all 10 predictors (same as OLS).

As λ increases: variables “drop out” one by one.

At very large λ : no predictors (intercept only).

The **effective model size** is the number of nonzero $\hat{\beta}_j$. We tune λ to get the right balance.

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Ridge vs LASSO: which to use?

It depends on the true underlying model.

- **LASSO** excels when few predictors matter — the zeros give a sparse, interpretable model
- **Ridge** excels when many predictors contribute with small-to-moderate effects — shrinking all helps
- When predictors are **highly correlated**: Ridge tends to shrink them together; LASSO may pick one and zero the others (arbitrary)



- If **interpretability** matters (which variables matter?): LASSO's zeros help
- If **prediction** is the only goal: try both, compare CV performance
- If predictors are **correlated**: consider **elastic net** (blend of ridge and LASSO)

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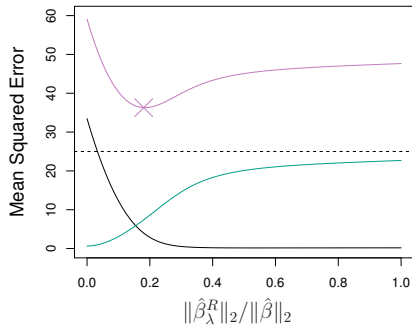
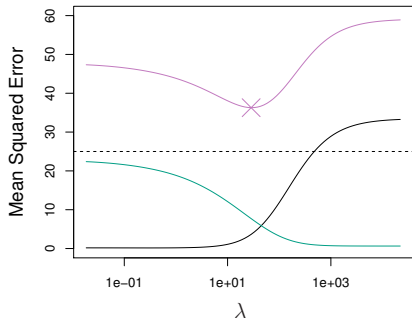
Choosing λ : cross-validation

1. Fit ridge/LASSO over a grid of λ values
2. For each λ , compute cross-validated MSE (e.g., 10-fold CV)
3. Pick λ that minimizes CV MSE

`cv.glmnet` does this automatically.



Choosing λ : the path



CV MSE (or test MSE) has a U-shape. We want the λ at the bottom.



One-standard-error rule

Sometimes we prefer a **simpler** model (fewer variables, more shrinkage) if its performance is “close enough.”

One-SE rule: Choose the largest λ (simplest model) whose CV MSE is within one standard error of the minimum.

In `cv.glmnet`: use `s = "lambda.1se"` instead of `s = "lambda.min"`.

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Combines ridge and LASSO:

$$\text{RSS} + \lambda \left[(1 - \alpha) \sum_{j=1}^p \beta_j^2 + \alpha \sum_{j=1}^p |\beta_j| \right]$$

with $0 \leq \alpha \leq 1$.

- $\alpha = 0$: pure ridge
- $\alpha = 1$: pure LASSO
- $\alpha = 0.5$: equal mix



When to use elastic net

Useful when predictors are correlated.

LASSO with correlated X 's: tends to pick one and zero the rest (arbitrary which one).

Ridge: shrinks correlated coefficients together.

Elastic net: gets some sparsity (from LASSO) plus grouped shrinkage (from ridge).

In `glmnet`: `alpha = 0.5` is a common default to try.

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glmnet: usage and cross-validation

```
library(glmnet)
X <- model.matrix(PhysiciansPer10000 ~ . - County, data = counties)
y <- counties$PhysiciansPer10000

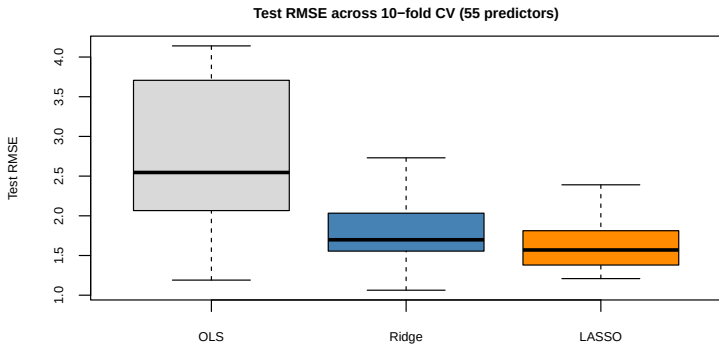
# Ridge (alpha=0), LASSO (alpha=1), elastic net (alpha=0.5)
fit_ridge <- glmnet(scale(X), y, alpha = 0)
fit_lasso <- glmnet(scale(X), y, alpha = 1)

# 10-fold CV to choose lambda
cv_lasso <- cv.glmnet(scale(X), y, alpha = 1, nfolds = 10)
cv_lasso$lambda.min    # minimizes CV MSE
cv_lasso$lambda.1se    # 1-SE rule (simpler model)
coef(cv_lasso, s = "lambda.min")
```

Comparing OLS, Ridge, and LASSO (physicians data)



Model: main effects + all pairwise interactions ($p = 55$: 10 main + 45 interactions)



Mean test RMSE (10-fold CV): OLS 2.72, Ridge 1.8, LASSO 1.64



- **Ridge** and **LASSO** shrink coefficients; LASSO can set some to exactly zero
- Both tune the bias-variance tradeoff via λ ; choose λ by cross-validation
- **Ridge**: all coefficients shrink, none to zero. Good when many predictors matter.
- **LASSO**: sparse solutions. Good when few predictors matter; aids interpretability.
- **Elastic net**: blend of both; useful when predictors are correlated
- Use `glmnet`; always center and scale. Consider the 1-SE rule for a simpler model.